

Bruce Quach

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EDUCATION

Louisiana State University, Baton Rouge, LA

Bachelor of Science, Computer Science

August 2020 - May 2024

GPA: 4.03/4.33

TECHNICAL SKILLS

- **Languages/Frameworks:** JavaScript (Node.js/Express.js), React.js/React Native, Redux, TypeScript, Python (Django, Flask), Java, SQL (MySQL, PostgreSQL), Tailwind, SCSS, D3.js, Cypress.io, HTML/CSS
- **Technologies:** AWS (S3, EC2, Cloudfront, Elastic Beanstalk), MongoDB, Docker, Redis, Firebase

PROFESSIONAL EXPERIENCE

Software Engineer II

June 2024 - Present

PMG - Dallas, TX

- Led frontend architecture for a rule-based table visualization tool that enables self-service data categorization via SQL, reducing analyst effort by 15+ hours/week and supporting \$2M+ in new client onboarding.
- Built scalable Node.js services with Cube.js and OpenAI's responses/assistant APIs to streamline SQL generation via natural language, simplifying Redshift, BigQuery, and Snowflake data retrieval for 60+ clients.
- Developed an AI agent-driven dashboard app architecting visualizations using Highcharts JS, speeding digital marketers' time to insights by 80% and reducing data analysts' workload by 20+ hours weekly.
- Optimized React.js and Redux apps to reduce digital marketing report and chart generation time from ~8 seconds to ~4 seconds, achieving a 50% performance improvement for 10,000+ daily report requests.
- Redesigned dashboard applications with TypeScript and custom design systems, increasing user adoption 5x.

Software Engineering Intern

November 2023 - August 2024

Timbalier Resources - Baton Rouge, LA

- Architected React/TypeScript web dashboards with Bootstrap components, Highcharts visualizations, ArcGIS, and FEMA's web APIs, helping local and state governments save up to \$2 million in flood insurance premiums.
- Deployed Flask apps to AWS (Docker, EB, Cloudfront, and S3) to assess natural disaster risks in 100+ counties.
- Designed AWS/SQL database connections to cache community API data, decreasing load times by up to 90%.

Software Engineering Intern

May 2023 - August 2023

Tesla - Fremont, CA

- Enhanced React.js tools built with Ant Design and D3.js to streamline vehicle quality engineer workflow, boosting Craftsmanship team efficiency and a projected \$5 billion Cybertruck revenue.
- Designed Python ElementTree XML parsing to create Pandas DataFrames via Django endpoints, reducing vehicle design file download times from 15 minutes to 30 seconds (98% improvement) for graphical model construction.
- Improved application performance up to 80% with Redux.js to efficiently access PostgreSQL enterprise data.

Software Development Intern

August 2022 - December 2022

Sparq - Baton Rouge, LA

- Developed scalable Java/Spring Boot application endpoints, improving internal scheduling efficiency by 25%.
- Built data access layers with Spring Data JPA supporting 15,000+ user records across MySQL and H2 databases.

PROJECTS

[Project Proudme](#) (React.js, MongoDB, Express.js/Node.js, Material UI, Recharts)

December 2022 - May 2024

- Optimized React.js virtual journaling and data visualization for 800+ students, improving daily health tracking engagement by 60% in a \$110k youth obesity prevention program funded by the LA Board of Regents.
- Developed personalized health plans with OpenAI chat completions and interactive Recharts data visualizations.